

False Positive Catamenial Epilepsy Classification: A Simulation Study

Research Article

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Abstract

Objective: To quantify false-positive catamenial epilepsy (CE) classification when seizure and menstrual-cycle diaries are independent.

Methods: We simulated 100,000 synthetic participants for 36 months in healthy ovulatory and heterogeneously ovulatory menstruating-age cohorts. CHOCOLATES generated seizure diaries, and HORMONE-CYCLE version 0.3.0 independently generated menstrual and hormone diaries using separate deterministic random streams; diaries were aligned directly by calendar day without reordering. We applied Herzog phase definitions and pattern-specific seizure-frequency ratio thresholds across calendar, complete-cycle, and full-diary windows. A negative-binomial analysis was used as a model-concordant statistical calibration check.

Results: Three-month false-positive rates were 41.6% in the healthy ovulatory cohort and 50.9% in the heterogeneous cohort. In 36-month windows, CE was classified in 11.4% and 35.9%, respectively. In the heterogeneous cohort, the C1/C2 rate was 11.6%, whereas C3 occurred in 37.1% of applicable windows. C1/C2 negative-binomial rates were 4.3% and 4.3%; the exploratory C3 model was positive in 6.1% of classifiable audit participants (95% confidence interval, 3.5%–10.3%).

Significance: False-positive CE classification is common when independent seizure and menstrual-cycle simulations are overlaid, particularly for heterogeneous cycles when C3 logic is used. Population, observation duration, phase definitions, and minimum-data rules materially affect results and should be reported explicitly.

Keywords: catamenial epilepsy; false-positive classification; menstrual cycle; seizure diary; simulation

Key Points

- Despite independence, 36-month Herzog false-positive rates were 11.4% in healthy-ovulatory and 35.9% in heterogeneous cohorts.
- The heterogeneous-cohort excess was primarily C3/inadequate-luteal-phase driven; full-diary C1/C2 rates were similar between cohorts.
- Three-month observation windows produced substantially higher false-positive classification than longer monitoring.
- Report the phase, its temporal definition, the comparison method, and prespecified minimum-data requirements.
- ~~Pattern-specific reporting, transparent denominators, and prespecified minimum-data and phase-label rules are essential.~~

Introduction

Catamenial epilepsy (CE) describes seizure exacerbation in relation to the menstrual cycle. Perimenstrual (C1), periovulatory (C2), and inadequate-luteal-phase (C3) patterns are ~~clinically recognized~~ the most often reported since formulated by Herzog and colleagues¹, ~~but~~ However, reported prevalence of CE varies with phase definitions, ratio thresholds, diary duration, and eligibility rules. Herzog and colleagues reported prevalence of 39.1% using three complete cycles and a two-of-three-cycle rule, while 44.2% was reported in the NIH Progesterone Treatment Trial.¹

~~Prospective diary classifications may disagree with participants' prior reports of catamenial seizure exacerbation. Self-report of CE is may be less unreliable when compared with prospective seizure diaries³. Such discordance can~~ This may reflect unrecognized diary-defined patterns, incomplete reporting of seizures, perceived patterns ~~not confirmed by the selected quantitative criterion~~ that are in fact spurious correlation, or instability of the pattern across limited observation windows. Both methods may be valid ~~in~~ under the right conditions.³

Seizures are sometimes temporally structured through clustering, circadian rhythms, and multidien rhythms. When these processes are independent of menstrual cycling, time windows can still align by chance. ~~We would consider~~ Such chance occurrences ~~to be represent CE misclassification or false-positive classifications~~ of CE. Ratio

definitions are especially vulnerable with sparse events: a small comparator count can yield a large or undefined ratio.⁴⁻⁶

We conducted a simulation with diaries and hormone cycles generated independently to estimate false-positive CE classification across definitions and observation windows. The study evaluates specificity behavior only. Our aim was to determine the expected “noise” produced by completely independent systems overlapping by chance.

Determining the rate of false-positive classification of CE under independence will contribute improved understanding of the state of the field of CE research and diagnosis. While the study will not provide an estimate of population prevalence of CE, it will estimate the contribution of false-positive CE classification to current commonly reported CE rates. Such research is important because patients with epilepsy have treatments modified, added, or withheld based on beliefs around the presence or absence of CE. This work is expected to contribute to research into reliable identification of CE as a phenomenon and ultimately better, evidence-based, epilepsy treatment decisions.

Methods

Study design

For each synthetic participant, seizure and menstrual/hormone diaries were generated independently for 36 months and aligned directly by calendar-day index. HORMONE-CYCLE generated the first menstrual cycle in full and selected diary day 1 uniformly from that cycle’s realized days; generation then continued forward without wrapping. Analyses were stratified by cohort and phase-labeling mode.

Cohorts

Two cohorts were used: a healthy cohort and a heterogeneous cohort. The healthy ovulatory cohort included 50,000 synthetic participants aged 18–45 years with regular ovulatory cycling-~~enforced~~. The heterogeneous menstruating-age cohort included 50,000 synthetic participants aged 13–55 years and allowed anovulation and irregularity. Rates of polycystic ovary syndrome, peri-menarche, perimenopause, and dysmenorrhea were sampled based on data-driven population estimates (Appendix S1).

Diary simulators

CHOCOLATES generated realistic self-reported seizure counts for all synthetic patients. The tool uses a hierarchical gamma–Poisson (equivalent to negative binomial) process with participant heterogeneity, L-relationship, clustering, multidien cycles, and ~~inter-seizure constraints~~maximum seizure rates.⁷ HORMONE-CYCLE version 0.3.0 is a custom simulator developed for the present project and released as an open-source community tool (see Appendix).⁸ It assigns age-specific participant variability components, samples right-skewed cycle realizations, and generates daily bleeding, ovulation, estradiol, and progesterone trajectories. Full hormone trajectories were retained for validation and the prespecified 1% daily audit sample; other paper-simulation participants used an RNG-equivalent compact renderer that retained identical cycle structure and inadequate-luteal-phase status without materializing unused concentration rows. Both simulators were designed in a data-driven manner to recapitulate statistical features observed within and across real patient populations, including randomized start times. The exact targets, sampling levels, realized distributions, and assumption-by-assumption rationale are reported in the Appendix.^{9-11 12}

As a separate aggregate cross-check not used for fitting, simulated 12-month age-stratified cycle means and participant-specific standard deviations were compared with a global period-tracking cohort.¹³ The comparison was prespecified as an external summary-level check rather than individual-level validation; Appendix S1 reports targets, margins, and the older-age discrepancy.

For ovulatory cycles, version 0.3.0 maps the complete daily LH-aligned serum estradiol and progesterone medians to the simulated ovulation event, uses PCHIP between those daily values, and preserves a broad luteal progesterone summit. Long follicular phases retain a terminal 14-day estradiol-maturation interval and use delayed-emergence or failed-wave geometry rather than stretching an ordinary-cycle template.^{9,14-16} An independent assay-specific serum cohort supplies subphase amplitude and ordering checks; Appendix S1 reports the numerical and visual validation

criteria. These checks concern population-level daily envelopes and do not establish individual clinical validity or reproduce within-day progesterone pulsatility.

Phase labeling and Herzog criteria

Strict Herzog phase labeling was the primary phase-definition scheme. The first day of a menstruation was labeled day 1. Menstrual phase comprised forward days 1–3 and backward days –3 to –1; follicular phase, forward days 4–9; ovulatory phase, forward days ≥ 10 through backward day –13; and luteal phase, backward days –12 to –4. Days outside these phases were unlabeled. A luteal-anchored sensitivity fixed the periovulatory interval at backward days –16 to –13 and assigned cycle-length variation to the follicular phase.¹⁷

Average daily seizure frequency (ADSF) was used for ratios. Windowed ratios were $C1 = \text{ADSF}(M)/\text{ADSF}(F+L)$ with threshold 1.69, $C2 = \text{ADSF}(O)/\text{ADSF}(F+L)$ with threshold 1.83, and $C3 = \text{ADSF}(O+L+M)/\text{ADSF}(F)$ with threshold 1.62 in applicable inadequate-luteal-phase cycles/windows. These exact distributional inflection values, rather than a rounded twofold rule, were used.¹⁷ A ratio was undefined when its numerator or comparator phase was absent or when both phase rates were zero; a positive numerator with zero comparator was treated as positive infinity.

C3 applicability

In the heterogeneous cohort, the adapter assigned a simulator-generated inadequate-luteal-phase indicator to all nonovulatory cycles and cycles without a valid ovulation day or progesterone field. For other ovulatory cycles, it used the maximum progesterone concentration on days 5–9 after ovulation, falling back to all postovulation days if that window was absent, and designated values < 5.0 ng/mL as inadequate. The C3 ratio then used only designated cycles. The 5.0-ng/mL cutoff follows the operational ovulatory criterion in the original Herzog study; implementation details and sensitivity limitations are in Appendix S1.¹⁷

CE Classification definitions

~~The exact Herzog 2004 cycle definition required exactly three complete 23–35-day cycles and the same positive pattern in at least two cycles.¹ In contrast, our windowed Herzog cycle definition pooled the phase-specific seizure-frequency ratios within each calendar, complete-cycle, or full-diary window. According to Herzog Exact Herzog cycle 2004 criteria required exactly three complete 23–35-day cycles and the same positive pattern in at least two eyes.⁴ C1/C2/C3 Windowed Herzog criteria pooled the phase-specific ratios in each calendar, cycle count, or full-diary window.~~ The minimum-data sensitivity required at least four calendar months or six complete cycles and at least four seizure days. These pragmatic thresholds represent an investigator-selected minimum-data sensitivity intended to reduce classifications dominated by extremely sparse event counts; they were not biologically derived. Nearby event thresholds were examined without searching for an optimal cutoff (Appendix S1). Cycle reproducibility required the same positive pattern in at least two-thirds of eligible cycles and a positive pooled ratio.

Model-concordant statistical calibration checks

Recurrent heterogeneous seizure counts are overdispersed, motivating negative binomial rather than simple Poisson models.⁷ The prespecified C1/C2 check used within-participant daily-count generalized linear models with log link, menstrual and ovulatory indicators, and cycle fixed effects where estimable. One-sided Wald tests were Holm-adjusted across C1 and C2; positivity required adjusted $P < .05$ and the corresponding Herzog rate-ratio threshold. Because CHOCOLATES uses a negative-binomial-like count process, this is a model-concordant statistical calibration check—not an alternative clinical diagnostic method or independent simulator validation.

A separately specified exploratory C3 model used the deterministic 1% daily audit sample. It restricted the heterogeneous cohort to complete cycles carrying the simulator-generated inadequate-luteal-phase indicator and contrasted combined O+L+M with F days using cycle fixed effects. Positivity required a one-sided $P < .05$ and rate ratio ≥ 1.62 . This standalone exploratory family was not pooled with C1/C2 and therefore received no cross-family multiplicity adjustment (Appendix S1).

Outcomes and statistics

The primary outcome was the participant-window false-positive classification rate among classifiable windows under independence. We also report rates among all attempted windows, explicit indeterminate reasons, and

mutually exclusive pattern categories. An indeterminate window was one in which an applicable rule could not return positive or negative because required phases, cycles, events, or denominator information were insufficient; non-applicability was kept separate. Definition-by-window-by-cohort estimates with fewer than 1,000 classifiable participant-windows were flagged as unstable. Wilson intervals quantify Monte Carlo uncertainty in a simulated proportion and do not include model-form uncertainty. Ten thousand simulated studies of 30, 50, or 100 participants sampled one random valid 3-month window per participant.

Ethics:

This study used only synthetic data. Institutional review board approval, patient consent, and clinical-trial registration were not applicable. All figures and tables are original; permission to reproduce third-party material was not applicable.

Results

Simulation cohort

The completed run included 100,000 synthetic participants. The healthy ovulatory cohort had 100.0% ovulatory cycles by design; the heterogeneous cohort had 79.0% ovulatory cycles, greater within-participant cycle-length variability, and similar seizure burden (Table 1). Every diary began at a randomly selected menstrual-cycle phase.

Full-diary CE classification

We used strict Herzog phase labeling and the prespecified windowed Herzog pattern-specific ratio thresholds across each participant's full 36-month diary. CE was classified in 11.4% of classifiable healthy-ovulatory participant-windows and in 35.9% of classifiable heterogeneous-cohort participant-windows under independence (Table 2). In the heterogeneous cohort, the C1/C2 union was 11.6%, whereas C3 was positive in 37.1% of C3-applicable windows. Thus, the between-cohort difference in the composite endpoint was primarily driven by the simulator-generated inadequate-luteal-phase designation. Because version 0.3.0 changes the progesterone envelope used to assign inadequate-luteal-phase status, all C3-dependent results were regenerated; in the retained 1% audit sample, the heterogeneous cohort's mean inadequate-luteal-phase cycle fraction was 22.8%.

~~In strict 23–25 day Herzog cycles 36-month windows, windowed Herzog C1/C2/C3 Herzog criteria thresholds classified 11.7% of healthy ovulatory and 36.0% of heterogeneous participants as having CE under independence (Table 2). In the heterogeneous cohort, the C1/C2 union was 11.7%, similar to the healthy cohort's 11.7%, whereas C3 positivity occurred in 37.0% of applicable windows. The excess was therefore primarily associated with the simulated inadequate-luteal-phase C3 mechanism (Figure 2 and Appendix S1).~~

Observation windows

Three-month false-positive rates were 41.6% and 50.9% in the healthy and heterogeneous cohorts, respectively. Rates declined with longer monitoring but remained definition-dependent (Figure 1). Exact Herzog applied to three complete cycles yielded 50.5% and 52.2% among classifiable windows, while many attempted windows were indeterminate. Appendix S1 reports C3 across every saved calendar and complete-cycle duration, minimum-data threshold sensitivities, and cumulative simulated C1, C2, and C3 ratio distributions.

Calibration checks

The full-diary C1/C2 negative-binomial false-positive rate was 4.3% in the healthy cohort and 4.3% in the heterogeneous cohort, close to the prespecified 5% Type I error rate. In the 1% daily audit sample, 427 of 500 heterogeneous participants had a C3-applicable ratio window, but only 197 met the exploratory C3 model's four-complete-inadequate-luteal-phase-cycle and four-seizure-day requirements. 12 participants were positive (6.1%; 95% Wilson confidence interval, 3.5%–10.3%; 2.4% of all 500 attempted participants).

Study-level and feature summaries

Across 10,000 simulated 3-month studies, apparent prevalence varied widely, particularly at $n=30$ (Table 3 and Figure 3). Associations with seizure burden, cycle variability, ovulatory fraction, age, and configured modifiers are reported in Appendix S1 as relationships between simulator inputs or realizations and simulated classifications, not clinical risk factors.

Discussion

This study quantified the false positive classification of several CE definitions using simulations that could not possibly include CE. Using pooled ratio criteria in random three-calendar-month windows, false-positive classification occurred in 41.6% of healthy-ovulatory and 50.9% of heterogeneous-cohort participant-windows. These results demonstrate the sensitivity of ratio-based classification to short observation periods. They should not be compared directly with published CE prevalence estimates^{1 2} of 39.1% and 44.2%, because those estimates were obtained in clinical cohorts using different cycle-eligibility, reproducibility, and denominator rules.

The analysis most closely aligned with the original Herzog 2004 procedure applied the same-pattern-in-two-of-three rule to exactly three complete 23–35-day cycles. Under that definition, false-positive classification occurred in 50.5% and 52.2% of classifiable windows in the healthy and heterogeneous cohorts, respectively, although many attempted windows were indeterminate. Even this comparison should be interpreted cautiously because the simulation represents complete independence and does not reproduce the mixture of true hormone-associated and unrelated seizure patterns present in clinical cohorts.

Accordingly, comparisons between simulated false-positive rates and empirical CE prevalence should use matched definitions, observation periods, eligibility requirements, reproducibility rules, and denominators. Numerical similarity alone does not establish how much of the reported clinical prevalence is attributable to false-positive classification.

This simulation study simulated of patients with seizures that were completely independent of their hormone cycles in order to determine false-positive rates of found a significant degree of CE misclassification that may also be occurring in the real world. For the most commonly used 3-month menstrual cycle diary and Herzog¹ ratios the false-positive rate in our simulation is, remarkably close to the population prevalence of CE (39–44%^{1,2}). We found the magnitude of such false positives depends on CE definition, observation length, analyzability rules, and menstrual-cycle assumptions.

The largest heterogeneous-cohort excess was C3-driven, whereas C1/C2 behavior was similar between cohorts.

The relationship between the simulated rate of CE and population prevalence from 3-month seizure diaries raises the question of whether C1/C2/C3 as reported in the literature are anything other than statistical artefact. In support of the Herzog classification and definition of CE, the simulations do not perfectly overlay onto empirical and real-world relationships between the menstrual cycle and seizures. For example, when degree of C1 exacerbation is taken into account for (Supplementary Table S4) the false-positive rate underestimates prevalence.

Across cumulative C1 ratio thresholds from 1.69 to 10, the proportions reported by Herzog (2015)² exceeded the corresponding simulated heterogeneous-cohort proportions by 1.01–10.81 percentage points, equivalent to relative differences of 19%–87% (Appendix Table S4).

There is a considerable body of literature on how menstrual cycle steroidal fluctuations can physiologically exacerbate seizures². Further, while there are few randomized controlled trials, there are studies reporting that treatments that stop the fluctuations of hormones may be effective in reducing seizure burden in women reporting CE. But overall, treatment evidence remains limited and mixed^{17 18,19}

This simulation study suggests actual rates of C1/C2/C3 CE as diagnosed using the Herzog criteria^{1 2} may be considerably less than current accounts. Furthermore, that the current commonly used definitions are vulnerable to statistical artefact. As such, The model-concordant negative binomial findings address implementation-level calibration under related model assumptions, not clinical usefulness or superiority. C1/C2 rates near 5% are expected for tests constructed around a 5% Type I error rate; ratio rules have no analogous guarantee. The exploratory C3 estimate was imprecise and used stricter eligibility in a 1% audit sample, so it should not be read as a direct replacement for the ratio-based C3 result. Its purpose is to show that a separately specified C3 contrast can be calibrated without implying independent validation of the simulator.

Earlier studies established biologically plausible and clinically important C1, C2, and C3 patterns, and treatment evidence remains limited and mixed.^{13-14,15} The present study instead estimates a diagnostic background rate

~~generated by chance alignment of independently simulated structured processes— this study points to It does not estimate true prevalence, sensitivity, positive predictive value, or treatment response. Rather, consider simulations of this sort show that as a lower limit that these CE commonly used criteria definitions are capable of—they are always expected to detect some degree of CE even when in reality none is present.~~

~~Several design implications for future research into CEHow.~~ First, short-window ratio classifications require caution when seizure counts are sparse. Second, C1, C2, and C3 should be reported separately. Third, classifiable and all-attempted denominators should be shown together. Fourth, study comparisons should match the original design: the Herzog^{1 2} 2004-benchmark is aligned with the exact three-complete-cycle rule, not a random three-calendar-month window.

~~Future empirical studies into CE and treatments ought to take into accountconsider the risk of false-positive in considering power calculations and group assignment.~~ Future ~~work directions for the simulation work should includes~~ compar~~inge~~ these simulation-based rates with empirical-null distributions derived from longitudinal paired seizure and menstrual diaries, using complementary preregistered within-person time shifts and matched cross-person pairing that preserve temporal structure, missingness, and participant-level dependence. Such work requires long-term real-world seizure and menstrual cycle diaries.

Strengths and limitations

Strict Herzog labeling was retained for historical comparability despite recognized limitations of fixed menstrual-phase assignment, particularly the expanding periovulatory interval in longer cycles. The luteal-anchored sensitivity therefore tested a fixed periovulatory definition.²⁰

The study is limited by simulator and adapter assumptions. CHOCOLATES and HORMONE-CYCLE were calibrated to published features, but summary-level calibration and the held-out aggregate cycle cross-check do not substitute for external validation using paired real diaries. Agreement was strongest at ages 18–45 and was more qualified after age 50 because the two large app cohorts reported different variability magnitudes. The configured heterogeneous-cohort modifiers are stress-test assumptions, the 5.0-ng/mL C3 rule operationalizes one historical definition, and no diary missingness was explicitly simulated (with the caveat that both simulators were developed to match self-reported diary statistical features). The hormone waveform is a daily population-median envelope rather than an individual endocrine time series; it omits within-day pulsatility, and the long-follicular failed-wave share is an investigator-set heterogeneity parameter rather than a published prevalence. Monte Carlo intervals do not include model-form uncertainty. Treatment changes and long-term seizure-rate trends were outside the scope of this paper and could invalidate stationarity assumptions or alter chance-alignment rates. The study does not evaluate false negatives or adjudicated seizure types.

~~Future work should compare these simulation-based rates with empirical null distributions derived from longitudinal paired seizure and menstrual diaries, using complementary preregistered within person time shifts and matched cross person pairing that preserve temporal structure, missingness, and participant level dependence. The model-concordant negative-binomial findings address implementation-level calibration under related model assumptions, not. These tests don't consider clinical usefulness or superiority. C1/C2 rates near 5% are expected for tests constructed around a 5% Type I error rate; ratio rules have no analogous guarantee. The exploratory C3 estimate was imprecise and used stricter eligibility in a 1% audit sample, so it should not be read as a direct replacement for the ratio-based C3 result. Its purpose is to show that a separately specified C3 contrast can be calibrated without implying independent validation of the simulator.~~

In conclusion, catamenial epilepsy misclassification ~~may ean~~ be common even when structured seizure and menstrual-cycle diaries are independent. Diagnostic and interventional studies should prespecify diary length, phase labeling, minimum-data thresholds, zero-denominator handling, ovulatory-status handling, and reproducibility criteria before interpreting apparent menstrual clustering as hormone-linked seizure exacerbation.

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Artificial Intelligence Use Disclosure

OpenAI Codex, a generative artificial intelligence coding and writing assistant, was used to assist with code review, software and notebook updates, consistency checking, and language editing. The authors retain responsibility for the accuracy, originality, and integrity of the work.

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Conflict of Interest

DMG reports funding from NIH K23NS124656, NIH R21NS142800, and the American Board of Psychiatry and Neurology. MBW is a co-founder of, serves as a scientific advisor and consultant to, and has a personal equity interest in Beacon Biosignals. The remaining authors have no conflicts of interest to disclose.

Author Contributions

Daniel M. Goldenholz: Conceptualization, methodology, software, formal analysis, visualization, writing—original draft, writing—review and editing, funding acquisition. Wesley T. Kerr: Conceptualization, methodology, validation, writing—review and editing. Sharon Chiang: Methodology, validation, writing—review and editing. M. Brandon Westover: Methodology, software, validation, writing—review and editing. Rachael L. Sumner: Methodology, validation, writing—review and editing.

Data and Code Availability

Code, configuration, derived tabular outputs, and the run manifest are available at <https://github.com/GoldenholzLab/catamenial-epilepsy-sim>. The completed primary simulation run used master seed 20260505; its manifest records configuration values, analysis-code and configuration fingerprints, file sizes, and SHA-256 checksums. The randomized-start analysis used `config_random_start_full.yaml`, and the Appendix S1 tables were derived from those primary outputs.

Ethical Publication Statement

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

Supporting Information

Appendix S1 contains the simulator source-to-parameter map, healthy-cycle calibration and held-out aggregate validation, hormone-waveform numerical and visual checks, representative hormone traces, assumption review, C3 algorithm, cumulative simulated Herzog-ratio panels, window and minimum-data sensitivities, exploratory C3 negative-binomial calibration check, pattern decomposition, feature associations, Appendix Tables A1–A8, Appendix Figures A1–A4, Tables S1–S8, and Figures S1–S5.

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Tables

Table 1. Simulated cohort characteristics.

Cohort	N	Age, y Mean	Age, y SD	Cycle length, d Mean	Cycle length, d SD	Ovulatory cycles	Seizure days/month	Seizures/month
Healthy ovulatory	50,000	31.5	7.8	28.75	3.93	100.0%	2.46	6.83
Heterogeneous menstruating- age	50,000	34.0	12.1	30.10	5.60	79.0%	2.45	6.80

Note. Values are cohort-level means from the completed run. Cycle-length SD is the mean within-participant cycle-length SD. N denotes simulated participants; y, years; d, days.

398 **Table 2. Participant-window false-positive results.**

Cohort	Analysis window	Definition	Classifiable / attempted	Classifiable, % (95% Wilson CI)	All attempted, %	Indeterminate, %
Healthy ovulatory	36-month full diary	Windowed Herzog thresholds	49,605 / 50,000	11.4 (11.1–11.6)	11.3	0.8
		Windowed Herzog C1/C2 union	49,605 / 50,000	11.4 (11.1–11.6)	11.3	0.8
		Herzog C1/C2 + ≥ 4 seizure-day minimum	48,359 / 50,000	10.3 (10.1–10.6)	10.0	3.3
		Negative-binomial C1/C2 calibration	48,359 / 50,000	4.3 (4.1–4.5)	4.2	3.3
	3-month calendar	Windowed Herzog thresholds	45,321 / 50,000	41.6 (41.2–42.1)	37.7	9.4
		Exact Herzog 2004, any pattern	22,113 / 50,000	50.5 (49.9–51.2)	22.3	55.8
		Windowed Herzog thresholds	49,587 / 50,000	35.9 (35.5–36.3)	35.6	0.8
		Windowed Herzog C1/C2 union	49,587 / 50,000	11.6 (11.3–11.9)	11.5	0.8
Heterogeneous menstruating-age	36-month full diary	Windowed Herzog C3 only	38,538 / 50,000	37.1 (36.6–37.6)	28.6	22.9
		Herzog C1/C2 + ≥ 4 seizure-day minimum	48,333 / 50,000	10.5 (10.2–10.8)	10.1	3.3
		Negative-binomial C1/C2 calibration	48,333 / 50,000	4.3 (4.1–4.4)	4.1	3.3
		Windowed Herzog thresholds	45,194 / 50,000	50.9 (50.5–51.4)	46.1	9.6
	3 complete cycles	Exact Herzog 2004, any pattern	18,164 / 50,000	52.2 (51.5–53.0)	19.0	63.7

Note. All rows use strict Herzog phase labeling. C3 is not applicable in the healthy ovulatory cohort, so the prior healthy C3 row is omitted. Indeterminate means that an applicable rule could not return positive or negative because required phases, cycles, events, or denominator information were insufficient; nonapplicability is distinct. Definition-by-window-by-cohort estimates based on fewer than 1,000 classifiable participant-windows were flagged as unstable.

405 **Table 3. Study-level Monte Carlo summaries for 3-month windows.**

Cohort	N/study	Definition	Mean apparent prevalence	2.5th–97.5th percentiles	Pr(apparent prevalence $\geq 39.1\%$)	Pr(apparent prevalence $\geq 44.2\%$)
Healthy ovulatory	30	Windowed Herzog thresholds	37.6%	20.0%–56.7%	45.9%	19.8%
	30	Windowed Herzog C1/C2 union	37.6%	20.0%–56.7%	45.9%	19.8%
	50	Windowed Herzog thresholds	37.8%	24.0%–52.0%	42.6%	14.7%
	50	Windowed Herzog C1/C2 union	37.8%	24.0%–52.0%	42.6%	14.7%
	100	Windowed Herzog thresholds	37.6%	28.0%–47.0%	34.6%	8.1%
	100	Windowed Herzog C1/C2 union	37.6%	28.0%–47.0%	34.6%	8.1%
Heterogeneous menstruating-age	30	Windowed Herzog thresholds	46.1%	30.0%–63.3%	79.9%	54.2%
	30	Windowed Herzog C1/C2 union	38.1%	20.0%–56.7%	48.3%	21.6%
	50	Windowed Herzog thresholds	46.1%	32.0%–60.0%	84.3%	56.1%
	50	Windowed Herzog C1/C2 union	38.1%	24.0%–52.0%	44.2%	16.1%
	100	Windowed Herzog thresholds	46.1%	36.0%–56.0%	90.4%	62.6%
	100	Windowed Herzog C1/C2 union	38.1%	29.0%–47.0%	38.4%	9.1%

406 **Note.** Each row summarizes 10,000 simulated studies using one random valid 3-month window per selected participant. Ranges are empirical
 407 percentiles across simulated studies, not confidence intervals. The 39.1% benchmark is aligned with the Herzog 2004 three-cycle/two-of-three rule;
 408 44.2% is the reported NIH Progesterone Trial proportion.
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Figure Legends and Figures

Figure 1. False-positive rates by length of observation window. Bars show the percentage of classifiable calendar-window participant-windows classified by strict Herzog phase labeling under independence. The dashed 5% line is a visual reference to the prespecified regression Type I error rate; it is not an expected target for the Herzog ratio rules.

Calendar-window false-positive rate by monitoring duration

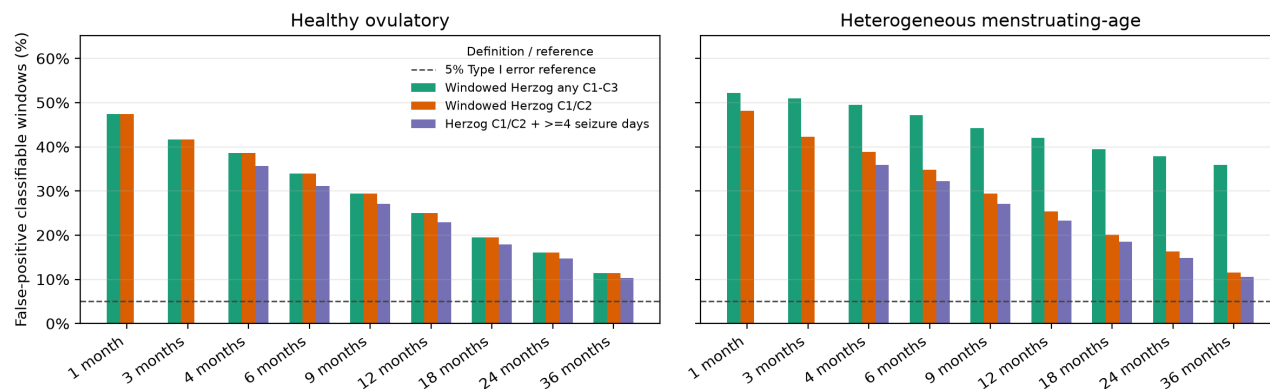


Figure 2. Pattern decomposition of false-positive catamenial classifications. Stacked bars show mutually exclusive participant-window categories in 36-month strict-Herzog windows. Percentages use all attempted windows. C3 is not applicable in the healthy ovulatory cohort; zero-height C3 segments there are not measured zero rates.

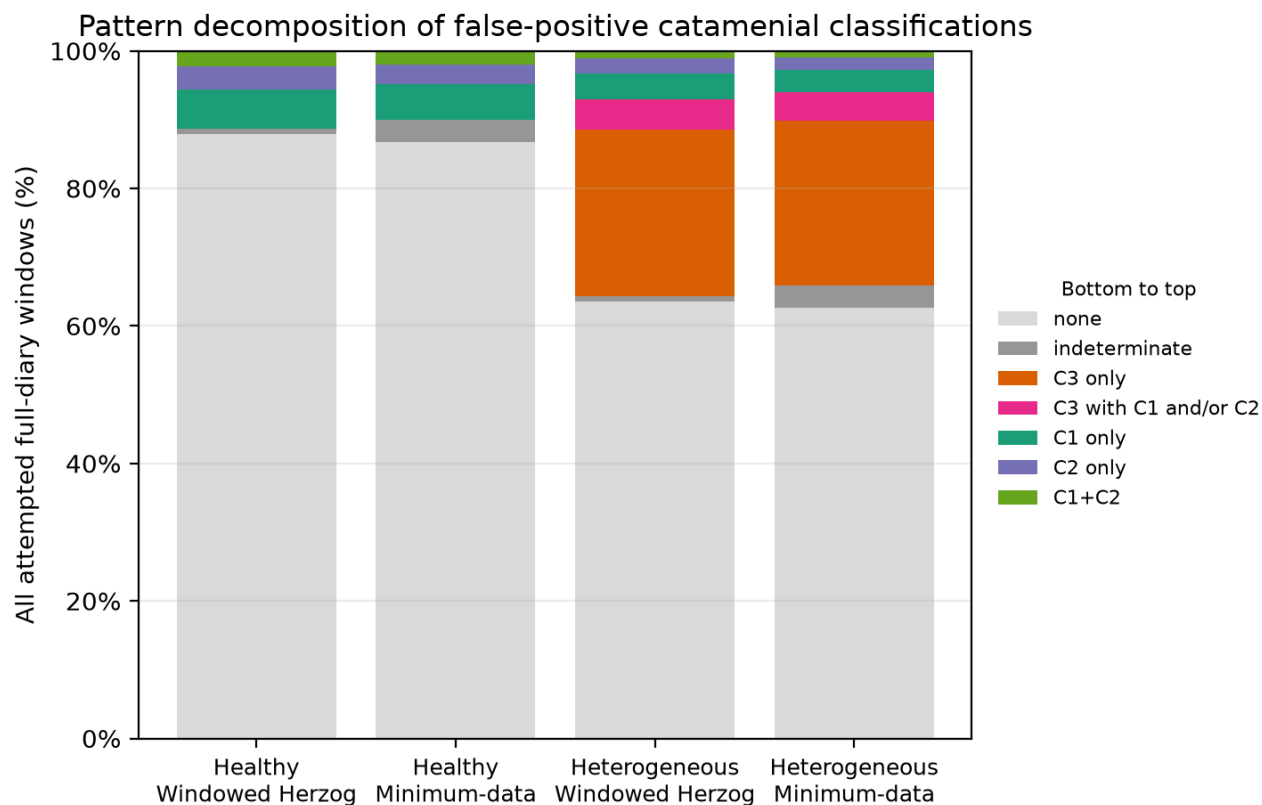


Figure 3. False-positive prevalence in simulated 3-month studies. Distributions show apparent prevalence in 10,000 studies using strict-Herzog windowed thresholds for $n=30$, 50, and 100. Vertical lines mark the historical 39.1% and 44.2% benchmarks.

